

Proposal: Alert Prediction Enhancements & Crime Data Research

Project Overview

This proposal outlines a two-part initiative: the enhancement of the existing Alert Prediction System and a broader crime data research effort. The upgrades aim to significantly improve the system's responsiveness, contextual accuracy, and adaptability across various user types (pedestrian, cyclist, driver), while the research will support intelligent alerting with deeper contextual insights derived from real-world data.

Scope of Work

1. Alert Prediction System Enhancements

1.1 Radius-Based Detection

Introduce a new API parameter allowing dynamic radius input (e.g., 5m, 10m). The system will return all hotspots detected within the user-defined area.

1.2 Vehicle Type Parameterization

Enhance the API with a `vehicle_type` input to distinguish between pedestrians, cyclists, and drivers. This allows for differentiated processing logic and more relevant alerts.

1.3 Historical Data Verification

Every detected hotspot will be checked against the last 6 months of incident data. If a location shows low recent activity, its danger score will be adjusted downward to reflect reduced risk.

1.4 Grid Cell Size Optimization by Context

Customize hotspot grid cell size based on vehicle type:

- Pedestrians: 5 meters
- Bicycles: 20 meters
- Cars: 50–100 meters

1.5 Caching & Performance Optimization

Introduce caching mechanisms for frequently accessed locations or high-traffic data, reducing API overhead and improving real-time responsiveness.

1.6 Containerization with Docker

Package the entire application in a Docker container to ensure easy deployment, scalability, and testing in diverse environments.

2. Crime Data Research

A research phase will be conducted to gather and analyze crime data from across the **United States and the United Kingdom**. Rather than limiting the scope to a few cities, this research will aim to cover as many regions as possible, depending on available open datasets and public data access points.

The objectives include:

- Identifying accessible and reliable data sources across the USA and UK
- Normalizing and aggregating crime data from different jurisdictions
- Analyzing crime trends and spatial hotspots by type and time
- Developing strategies for integrating these insights into the Alert Prediction System

Further strategies and data acquisition methods will be outlined once the data landscape is fully evaluated.

Deliverables

- Updated API with support for radius-based queries and vehicle type input
- Enhanced scoring logic using 6-month historical trend verification
- Adaptive grid sizing logic based on user context
- Optimized system performance through intelligent caching
- Dockerized application for portable deployment
- Crime data acquisition plan and insights applicable to alert logic

Timeline: 10–14 days

Total Cost: 300 USD

Please let me know if anything is unclear or if you'd like any modifications to the proposal. I'm happy to make adjustments based on your feedback.